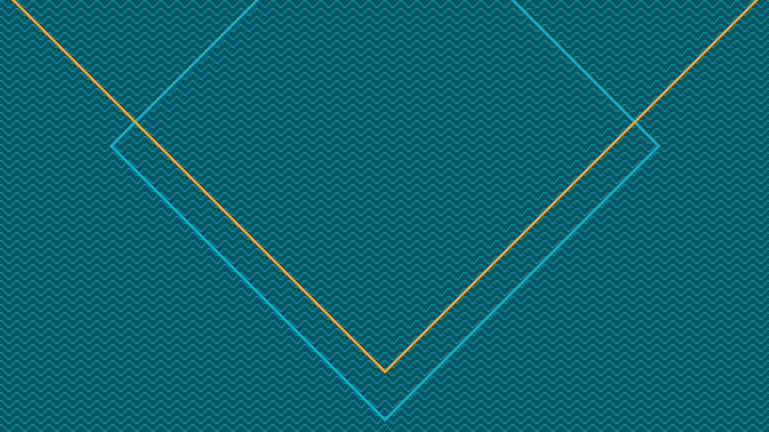


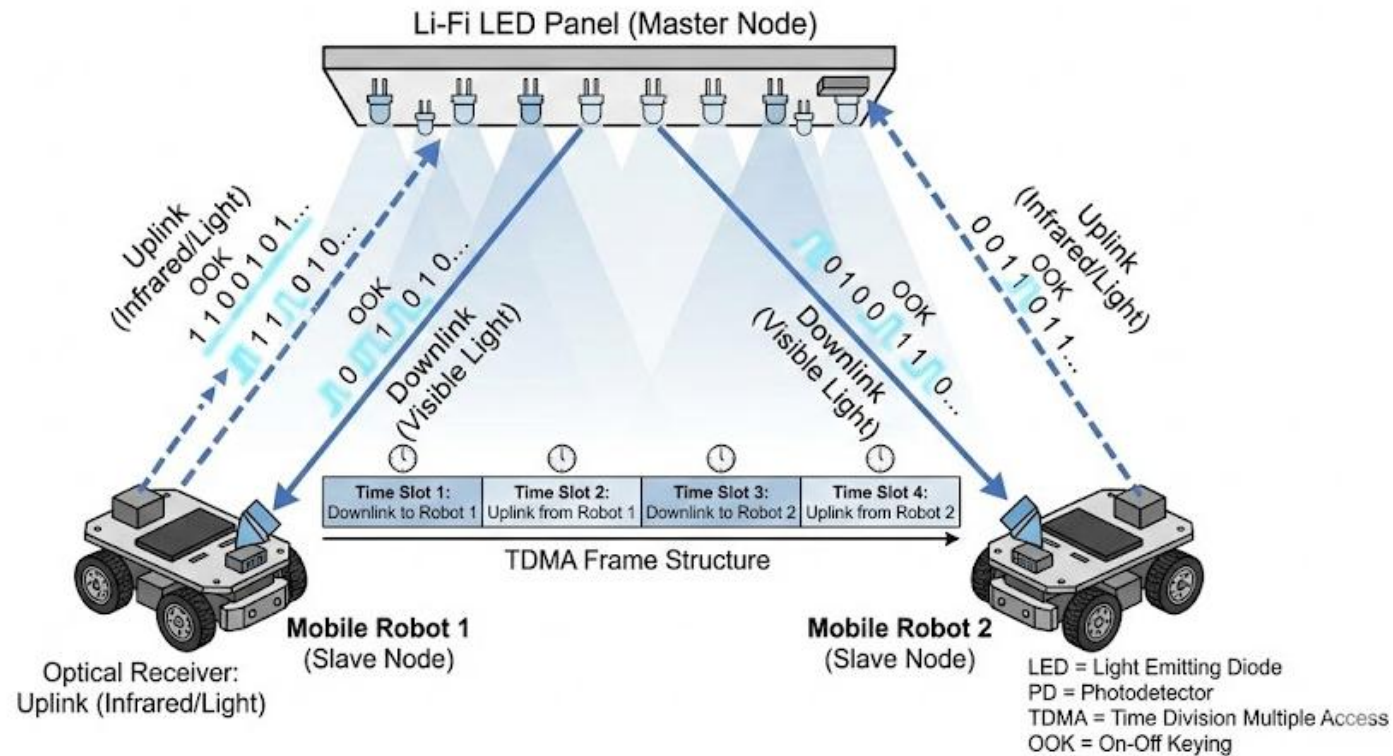
A decorative graphic at the top center of the slide, consisting of two overlapping diamond shapes. The outer diamond is formed by two orange lines meeting at a point at the bottom, and the inner diamond is formed by two teal lines meeting at a point at the bottom. The background is a solid teal color.

Final Year Project

Presentation

Project Title

Li-Fi Based Multi-Robot Communication System with Centralized Probabilistic Grid Mapping



Introduction

Wireless communication plays a crucial role in coordinated robotic systems. Conventional radio-frequency (RF) technologies such as Wi-Fi and Bluetooth suffer from spectrum congestion, electromagnetic interference, and security concerns, especially in RF-restricted environments. Li-Fi (Light Fidelity), which utilizes visible light for data transmission, has emerged as a promising alternative due to its high security, immunity to RF interference, and ability to reuse existing lighting infrastructure.

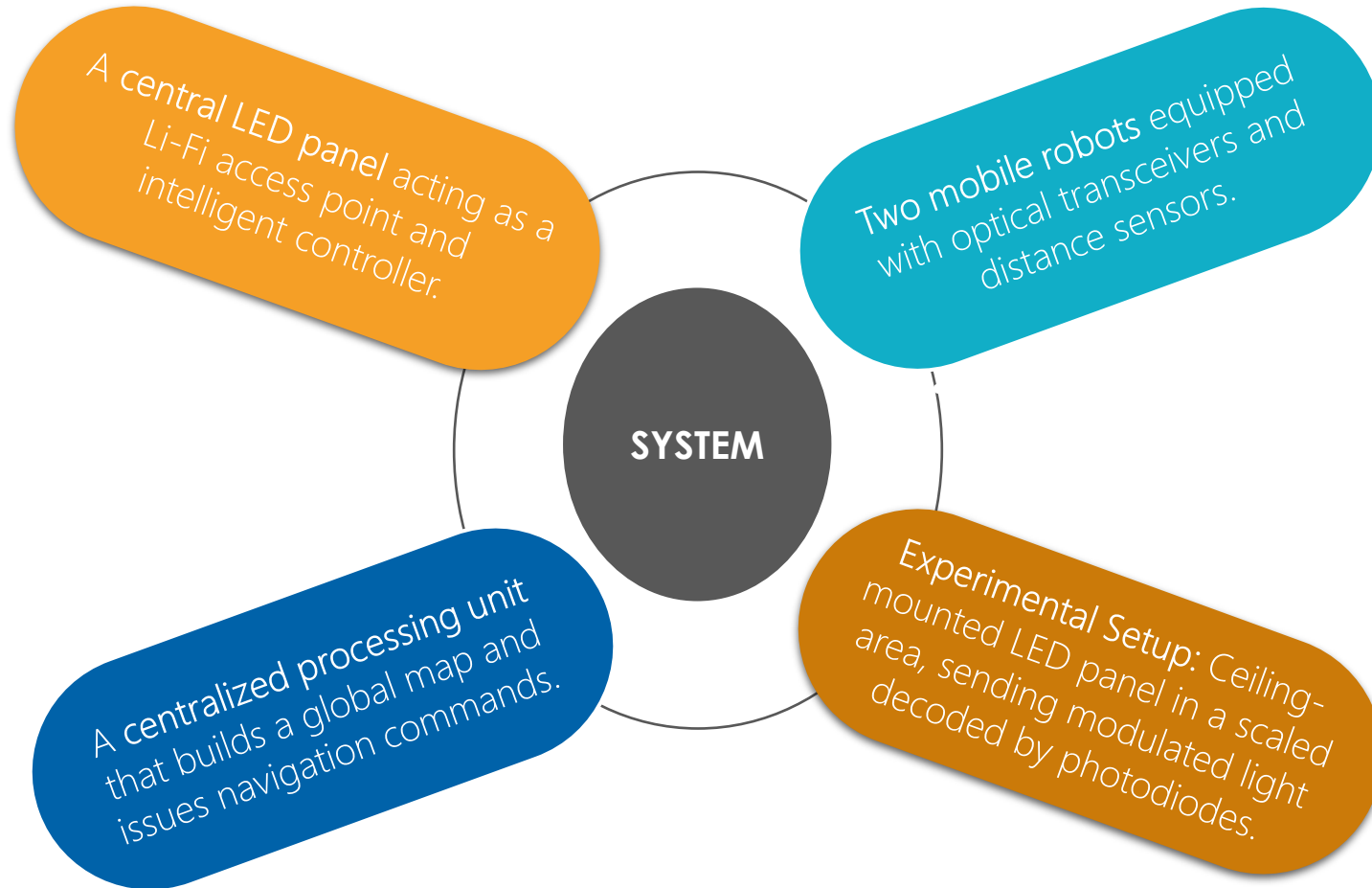
Introduction

- This project proposes the design and implementation of a **Li-Fi based multi-robot communication framework** in which two mobile robots communicate with a centralized controller through a visible light link. In addition to communication, the system integrates a **centralized intelligence module** that performs **probabilistic occupancy grid mapping** and **path planning**, thereby enabling coordinated navigation within a defined indoor environment.
- The project is demonstrated using a **scaled indoor model**, ensuring reliable operation with low-cost optical components while preserving the fundamental principles of Li-Fi communication.

Objectives

- To design a reliable Li-Fi communication link between a central LED panel and multiple mobile robots.
- To implement multi-node communication with robot addressing and collision-free data transmission.
- To develop a centralized probabilistic grid-based mapping system using sensor data collected by robots.
- To apply intelligent path-planning algorithms for efficient robot navigation.
- To analyze the impact of ambient light and noise on Li-Fi system performance.
- To demonstrate coordinated robot motion using visible light communication.

System Overview



The LED panel is mounted above the operating area in a scaled experimental setup, emulating a ceiling-mounted Li-Fi system. Communication is achieved through **intensity modulation of visible light**, and data is decoded using photodiodes and signal conditioning circuits.

System Architecture

1. Central Li-Fi Panel (Master Node)

- Functions:
 - Downlink transmission of control and navigation commands.
 - Uplink reception of sensor and status data from robots.
 - Centralized map construction and decision making.
- Components:
 - High-power LED array
 - LED driver with modulation capability
 - High-sensitivity photodiode with focusing optics
 - Analog signal conditioning circuitry
 - Microcontroller / embedded processor

2. Mobile Robots (Slave Nodes)

- Functions:
 - Receive motion commands from the panel.
 - Sense obstacles in the environment
 - Transmit sensor readings back to the panel
 - Execute centrally planned paths
- Components:
 - Upward-facing photodiode for downlink reception
 - LED transmitter for uplink communication
 - Microcontroller for control and decoding
 - Ultrasonic or infrared distance sensor
 - Motor driver and DC motors

Communication Framework

1. Network Topology

A **centralized star topology** is used, where all communication is coordinated by the LED panel. Robot-to-robot communication is performed indirectly through the central controller.

2. Communication Method

- Intensity modulation using On-Off Keying (OOK) or PWM
- Low-data-rate digital packets
- Robust and noise-tolerant design

Communication Framework

3. Medium Access Control

Time Division Multiple Access (TDMA) is employed to:

- Avoid data collisions
- Enable deterministic communication
- Simplify synchronization

4. Packet Structure

Each packet contains:

- Start delimiter
- Robot ID
- Command or sensor data
- Error detection field (checksum)

Centralized Intelligence and Mapping

1. Occupancy Grid Mapping

The environment is represented as a two-dimensional grid, where each cell stores the probability of being occupied. Instead of binary occupancy, a probabilistic approach is used to handle sensor uncertainty.

Each sensor observation updates cell occupancy using a simplified Bayesian update method, gradually refining the map as robots explore the area.

2. Path Planning

Using the constructed grid map, the central controller computes optimal paths using the **A*** algorithm. The algorithm selects collision-free paths while minimizing travel distance.

Planned paths are transmitted to robots as sequences of waypoints, reducing onboard computational requirements.

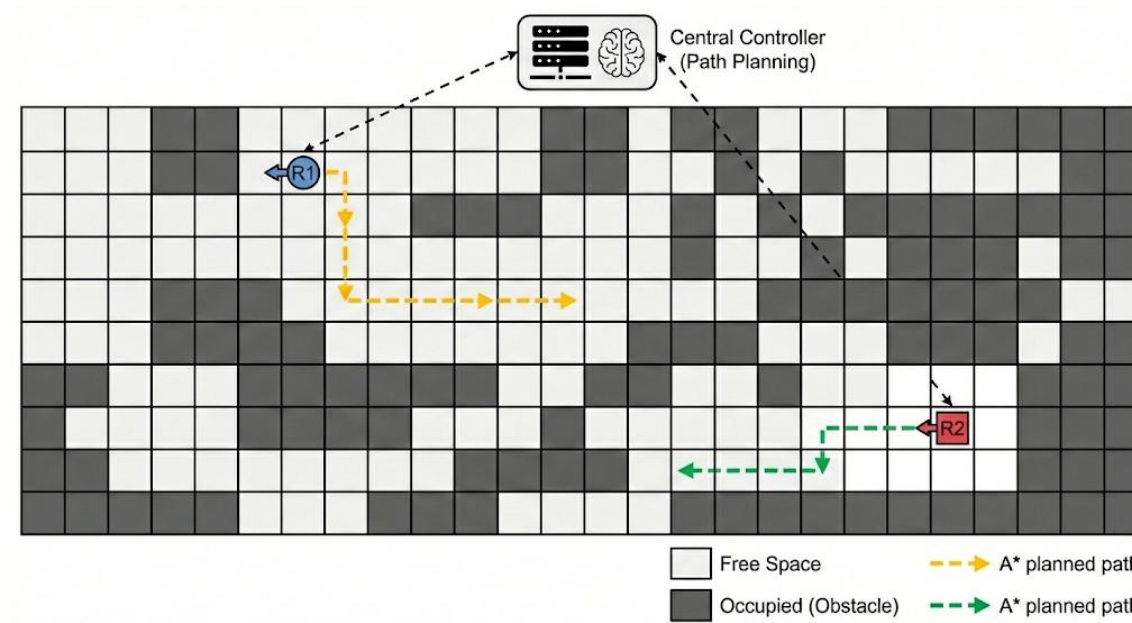
Centralized Intelligence and Mapping

1. Decision-Making Strategy

The intelligence layer includes:

- Obstacle avoidance logic
- Goal-directed navigation
- Simple rule-based decision making

This approach provides meaningful intelligence without introducing excessive computational complexity.



Experimental Setup

The system is demonstrated in a **scaled indoor environment** that emulates a ceiling-mounted Li-Fi system. This setup ensures reliable communication using low-cost components while maintaining realistic optical geometry.

Measurements include:

- Received signal strength vs distance
- Bit error rate under different lighting conditions
- Communication latency
- Mapping accuracy

Scope and Limitations

Scope:

- Indoor, line-of-sight environments
- Low-speed data transmission
- Centralized intelligence

Limitations:

- Sensitivity to ambient light
- Limited operating area
- Scaled demonstration environment

Applications

- RF-Restricted Environments: Hospitals, aircraft cabins, or sensitive industrial areas.
- Smart Indoor Robotics: Coordinated robot navigation for warehouses or labs.
- Secure Communication Systems: Data transfer with minimal interference risk.
- Educational & Research Platforms: Demonstrates Li-Fi and multi-robot coordination.
- Prototype for IoT-Integrated Automation: Shows potential for smart building applications.



Thank You